

PERSONAL FINANCE TRACKER

A MINI-PROJECT REPORT

Submitted by

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in partial fulfillment of the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

An Autonomous Institute

NOVEMBER 2025

BONAFIDE CERTIFICATE

Certified that this project “**PERSONAL FINANCE TRACKER**” is the bonafide work of “**DANYA A**” who carried out the project work under my supervision.

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This mini project report is submitted for the viva voce examination to be held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

A personal finance tracker is a digital tool designed to help individuals manage their money more effectively. It enables users to record and monitor income, expenses, savings, and investments, offering a clear overview of their financial health. By analysing spending patterns and setting budget goals, the tracker promotes financial discipline and informed decision-making. It empowers users to take control of their finances, reduce unnecessary expenditures, and work toward long-term financial stability.

ACKNOWLEDGEMENT

We express our sincere thanks to our beloved and honorable chairman **MR. S. MEGANATHAN** and the Chairperson **DR. M. THANGAM MEGANATHAN** for their timely support and encouragement.

We are greatly indebted to our respected and honorable principal Dr. **S.N. MURUGESAN** for his able support and guidance.

No words of gratitude will suffice for the unquestioning support extended to us by our Head of the Department **Dr. J.M. GNANASEKAR** for being ever supporting force during our project work

We also extend our sincere and hearty thanks to our internal guide **Dr. E.K. SUBRAMANIAN**, for his valuable guidance and motivation during the completion of this project.

Our sincere thanks to our family members, friends and other staff members of Artificial Intelligence and Data Science.

DANYA A

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Managing personal finances is essential, yet many people struggle to track their income and expenses. A **Personal Finance Tracker** helps users monitor their financial health by recording income, tracking expenses, and analyzing spending habits. This project aims to develop a system with a **Java-based front end** and a **DBMS backend** to help users manage finances efficiently and make informed decisions.

1.2 SCOPE OF THE WORK

- Developing a **user-friendly Java front-end** for data entry and visualization.
- Implementing a **DBMS backend** to securely store income and expense records.
- Categorizing income and expenses into different predefined or user-defined categories.
- Calculating **total income, total expenses, and net balance** automatically.
- Generating **warnings or alerts** if expenses exceed a defined threshold.
- Providing **basic financial analysis** to help users identify spending patterns.
- Supporting future enhancements like graphical reports, budget planning, or mobile integration.

1.3 PROBLEM STATEMENT

- Record incomes and expenses from multiple categories.
 - Compute total income, total expenditure, and current balance.
 - Alert users in case of overspending.
- The absence of such a system leads to poor financial decisions, difficulty in saving, and increased financial stress.

1.4 AIM AND OBJECTIVE OF THE PROJECT

The aim of this project is to develop a Personal Finance Tracker that enables users to manage and analyze their income and expenses effectively. It uses a Java-based front end and a DBMS backend to record, categorize, and summarize financial transactions.

CHAPTER 2

SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	:	Intel i5
Memory Size	:	8GB (Minimum)
HDD	:	1 TB (Minimum)

2.2 SOFTWARE SPECIFICATIONS

Operating System	:	WINDOWS 10
Front – End	:	Java
Back - End	:	MySql
Language	:	Java, MySQL

CHAPTER 3

MODULE DESCRIPTION

1. Main Application Module (`MainApp`)

Purpose: Acts as the user interface for the console application.

Responsibilities:

- Present the initial menu for registration and login.
- Navigate the user to different functionalities: adding categories, adding transactions, viewing reports, or exiting.

Inputs: Username, password, email, category details, transaction details.

Outputs: Console messages for registration/login success, transaction status, and financial report.

2. User Management Module (`UserService`)

Purpose: Handle user registration and login functionality.

Responsibilities:

- Register new users in the `users` table.
- Authenticate users during login by verifying username and password.

Inputs: Username, password, email.

Outputs: User ID for successful login or -1 for failure.

Note: Passwords are stored in plaintext in this implementation. For real-world applications, **passwords should be hashed** for security.

3. Finance Management Module (`FinanceService`)

Purpose: Manage categories, transactions, and reports.

Sub-modules & Responsibilities:

1. Category Management

- Add a new category (income or expense) to the `categories` table.
- Associate the category with the logged-in user.

2. **Transaction Management**

- Add income or expense transactions linked to a category.
- Store transactions in the `transactions` table.

3. **Report Generation**

- Calculate total income and total expenses for a user.
- Display a summary report and warn if expenses exceed income.

Inputs: Category name, type, transaction amount, category ID, description.

Outputs: Success/error messages, financial report with totals and warnings

CHAPTER 4

SAMPLE CODING

```
import java.util.*;

class Category {

    String name, type;

    Category(String name, String type) { this.name = name; this.type = type; }

}

class Transaction {

    Category category;

    double amount;

    String description;

    Transaction(Category category, double amount, String description) {

        this.category = category; this.amount = amount; this.description = description;

    }

}

public class MiniFinanceApp {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        List<Category> categories = new ArrayList<>();
```

```
List<Transaction> transactions = new ArrayList<>()

System.out.println("=== Mini Finance Tracker ===");

System.out.print("Enter username: ");

String username = sc.nextLine();

System.out.print("Enter password: ");

String password = sc.nextLine();

System.out.println("Login successful! Welcome, " + username + "\n");


while (true) {

    System.out.println("1. Add Category  2. Add Transaction  3. Show Report  4.
Exit");

    int option = sc.nextInt(); sc.nextLine();


    switch(option) {

        case 1:

            System.out.print("Category name: ");

            String cname = sc.nextLine();

            System.out.print("Type (income/expense): ");

            String ctype = sc.nextLine();

            categories.add(new Category(cname, ctype));

            System.out.println("Category added!\n");
```

```

        break;

    case 2:

        if(categories.isEmpty()) {

            System.out.println("Add a category first!\n");

            break;

        }

        System.out.println("Select category:");

        for(int i=0;i<categories.size();i++)

            System.out.println((i+1)+"."+categories.get(i).name+"
("+categories.get(i).type+""));

        int catIndex = sc.nextInt()-1; sc.nextLine();

        System.out.print("Amount: "); double amt = sc.nextDouble(); sc.nextLine();

        System.out.print("Description: "); String desc = sc.nextLine();

        transactions.add(new Transaction(categories.get(catIndex), amt, desc));

        System.out.println("Transaction added!\n");

        break;

    case 3:

        double totalIncome=0, totalExpense=0;

        for(Transaction t: transactions){

            if(t.category.type.equalsIgnoreCase("income"))    totalIncome    +=
t.amount;

```

```
        else totalExpense += t.amount;

    }

    System.out.println("--- Report ---");

    System.out.println("Total Income: "+totalIncome);

    System.out.println("Total Expense: "+totalExpense);

    if(totalExpense > totalIncome) System.out.println(" Overspent!");

    else System.out.println(" Within budget");

    System.out.println();

    break;

case 4:

    System.out.println("Goodbye!");

    sc.close();

    return;

default:

    System.out.println("Invalid option!\n");

}

}

}

}
```

Sample 1

Sample 1 depicts the display code, that gets the data from the database i.e. being stored there and represented on users demand with the layout and measurements i.e. being already specified.

```
import java.sql.*;

public class DisplaySample {

    public static void main(String[] args) {

        String url =
            "jdbc:mysql://localhost:3306/myproject?useSSL=false&serverTimezone=UTC";

        String user = "root";

        String password = "Danya@123";

        String sql = "SELECT category_name, type, amount FROM categories c "
            + "LEFT JOIN transactions t ON c.category_id = t.category_id";

        try (Connection conn = DriverManager.getConnection(url, user, password);

            Statement stmt = conn.createStatement();

            ResultSet rs = stmt.executeQuery(sql)) {
```

```
System.out.println("Category Name\tType\tAmount");

System.out.println("-----");

while (rs.next()) {

    String name = rs.getString("category_name");

    String type = rs.getString("type");

    double amount = rs.getDouble("amount");

    System.out.println(name + "\t" + type + "\t" + amount);

}

} catch (SQLException e) {

    System.out.println("Error fetching data: " + e.getMessage());

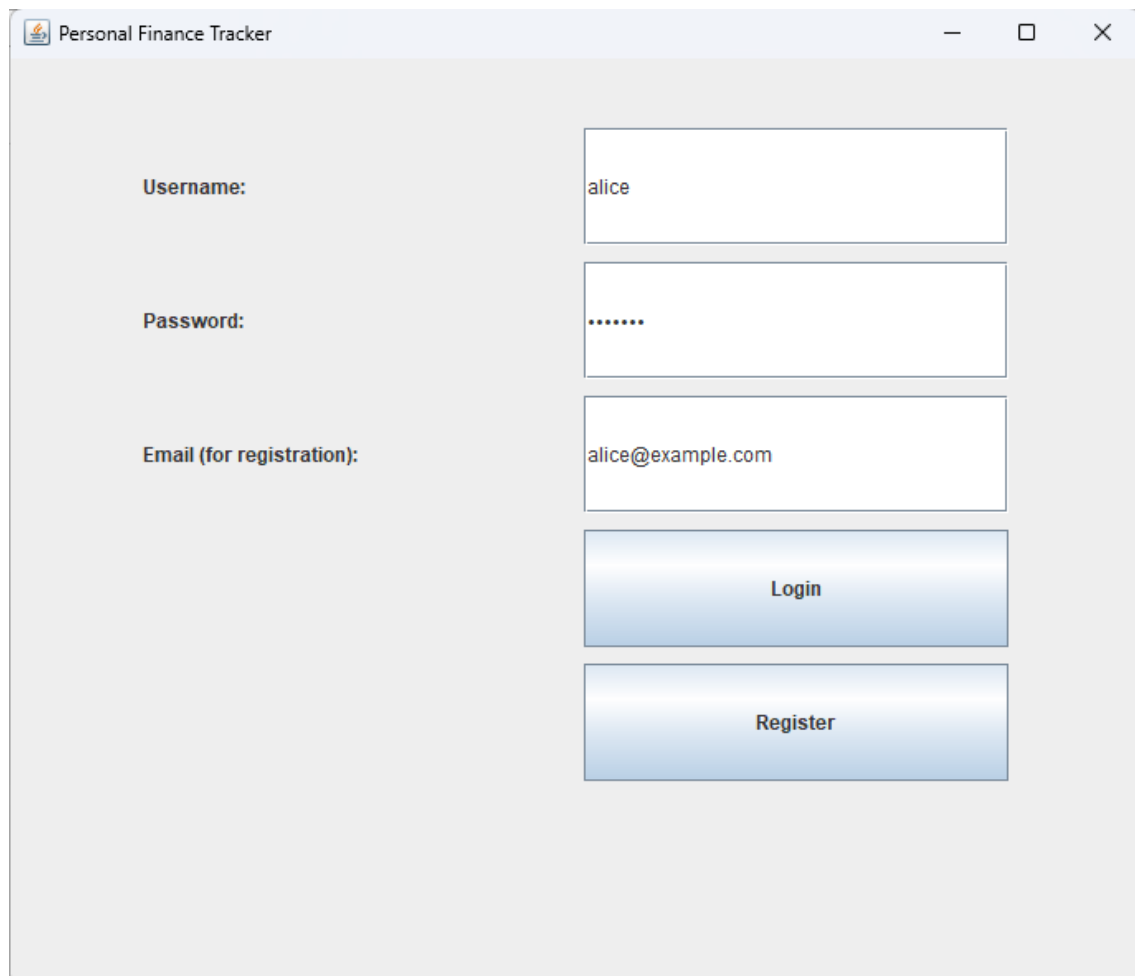
}}}
```

Sample 2

Sample 2 depicts the category and transaction of the code, where it displays income or expense details and enter user data and store it in database

CHAPTER 5

SCREEN SHOT



The screenshot shows a web application window titled "Personal Finance Tracker". The window contains a login and registration form. The form has three input fields: "Username:" with the value "alice", "Password:" with masked characters "*****", and "Email (for registration):" with the value "alice@example.com". Below the input fields are two buttons: "Login" and "Register".

Username:	alice
Password:	*****
Email (for registration):	alice@example.com
Login	
Register	

Fig 5.1 login/register page

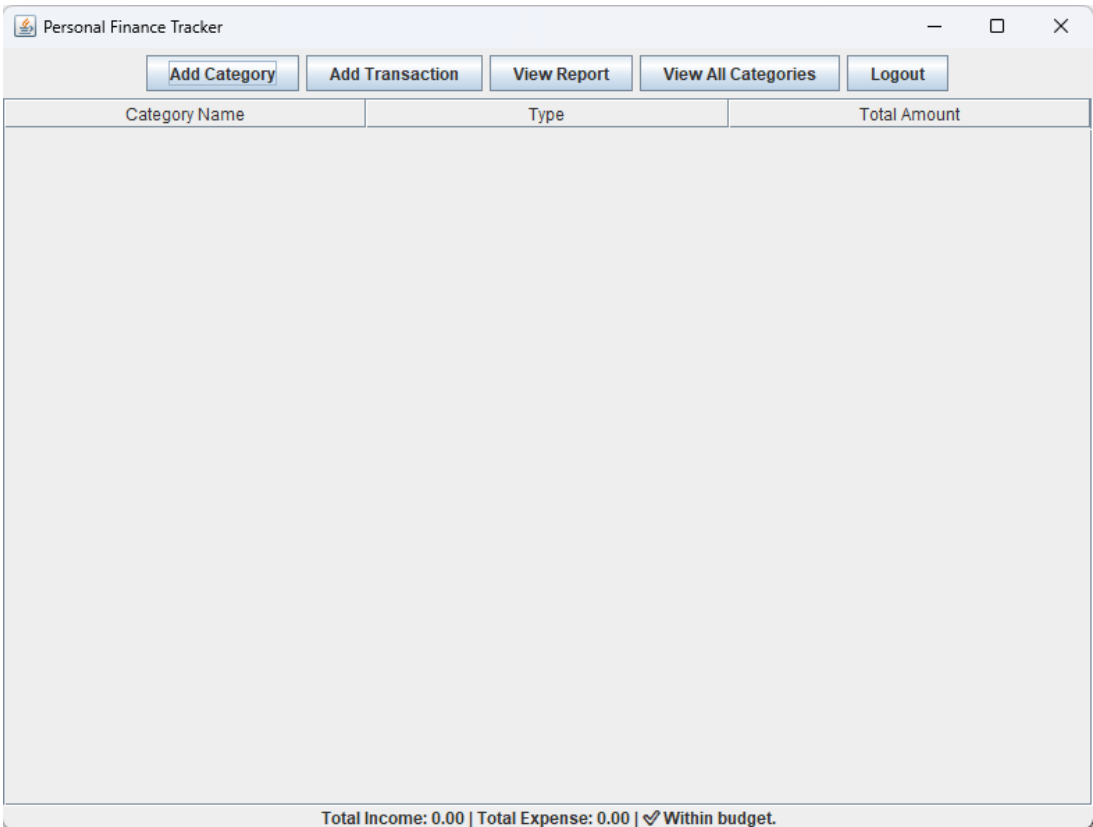


Fig 5.2 Introduction page

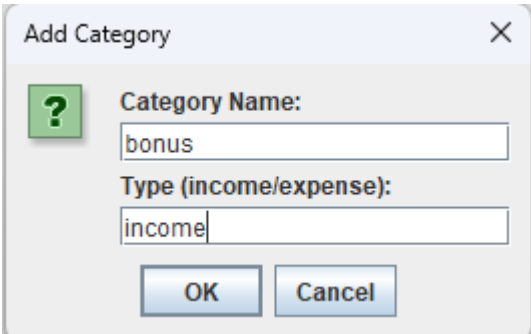


Fig 5.3 category details

?

Select Category:

bonus

Amount:

5000

Description:

income

OK

Cancel

Fig 5.4 Transaction details

Personal Finance Tracker

Add Category

Add Transaction

View Report

View All Categories

Logout

Category Name	Type	Total Amount
bonus	income	5000.0

Total Income: 10000.00 | Total Expense: 0.00 | ✓ Within budget.

Fig 5.5 Viewing reports

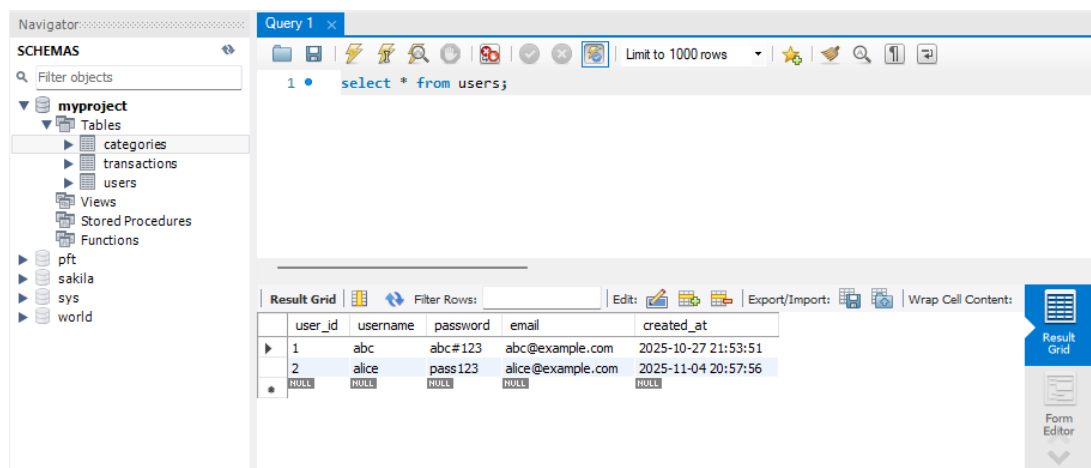


Fig 5.6 Database creation

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

The Personal Finance Tracker helps users manage their income and expenses by allowing them to create categories, add transactions, and view detailed reports. It provides a clear summary of spending and helps users stay within their budget. In the future, the app can be improved with features like charts and graphs, recurring transactions, budget alerts, data export, enhanced security, and mobile accessibility to make financial management even easier and more convenient.

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